Windows Delivery Optimization — Bandwidth Savings & Peer Content Distribution (Intune)

Implementation Solution Guide

Version 1.0.0 · 2026-07-30

Contents

Executive Summary — Windows Delivery Optimization (Bandwidth Savings & Peer Content Distribution) 5

1. Solution Overview — Windows Delivery Optimization (Bandwidth Savings & Peer Content Distribution) 5

- What Delivery Optimization is . . . 5
- Which content types Delivery Optimization can optimize . . . 6
- How it works — the peer-to-peer + cloud/HTTP model . . . 6
- Download Mode — the primary decision (values verified against Microsoft Learn) . . . 6
- Scoping peer groups correctly — GroupID / GroupIDSource . . . 7
- Bandwidth controls (verified names) . . . 7
- Cache controls (verified names) . . . 8
- Microsoft Connected Cache (MCC) — decision-level only . . . 8
- Monitoring & proving savings . . . 8
- Benefits . . . 9
- Limitations (be honest with buyers) . . . 9
- The standard deployment model . . . 9

2. Requirements Guide — Windows Delivery Optimization 9

- 2.1 Licensing (Intune) . . . 9
- 2.2 Supported Windows edition and version . . . 10
- 2.3 Delivery Optimization service running (DoSvc) . . . 10
- 2.4 Peer connectivity — firewall / port 7680 . . . 10
- 2.5 Network access to Delivery Optimization cloud + content endpoints . . . 10
- 2.6 A single, non-conflicting delivery configuration . . . 10
- 2.7 Pilot and Production device groups . . . 11
- 2.8 (Recommended) A Delivery Optimization measurement baseline . . . 11
- Checklist . . . 11

Global Requirements — Intune Solutions (category-level) 11

- G1. Microsoft Entra ID tenant . . . 11
- G2. Microsoft Intune subscription . . . 11
- G3. Administrative roles . . . 12
- G4. Device enrollment in Intune . . . 12
- G5. Pilot security group . . . 12
- G6. Supported Windows baseline . . . 12
- G7. Network access to Microsoft cloud endpoints . . . 12

3. Implementation Guide — Windows Delivery Optimization via the Intune Settings Catalog 12

Step 1 — Assess: capture the baseline . . . 13
Step 2 — Create the Delivery Optimization policy . . . 13
Step 3 — Set the Download Mode and scope the peer group . . . 13
Step 4 — Apply bandwidth controls . . . 14
Step 5 — Size the peer cache . . . 14
Step 6 — (Decision-level) point at Microsoft Connected Cache, if you have one . . . 15
Step 7 — Finish and assign to Pilot . . . 15
Step 8 — Validate on the Pilot (gate before expanding) . . . 15
Step 9 — Expand to Production . . . 16
3.9 US Government (GCC / GCC High / DoD) notes . . . 16
5. Testing & Validation — Delivery Optimization 16
Test 1 — Policy applied . . . 16
Test 2 — Delivery Optimization service healthy . . . 17
Test 3 — Peer connectivity (port 7680) . . . 17
Test 4 — Peering actually happens (the core proof) . . . 17
Test 5 — Group scoping is correct . . . 17
Test 6 — Bandwidth caps respected . . . 17
Test 7 — Savings measured vs. baseline . . . 17
Test 8 — HTTP fallback safety . . . 17
Exit criteria (all must be true before Production) . . . 18
6. Troubleshooting — Delivery Optimization 18
No peering — every device downloads over HTTP (BytesFromPeers = 0) . . . 18
Devices peer with the wrong set (group scoped incorrectly) . . . 18
Savings don't appear in reports . . . 19
Conflicting configuration / policy not applying . . . 19
Bandwidth caps not respected / foreground starved . . . 19
Bypass (100) side-effects on Windows 11 . . . 19
US Government specifics . . . 20
7. Best Practices — Delivery Optimization 20
Download mode & grouping . . . 20
Bandwidth . . . 20
Cache . . . 20
Microsoft Connected Cache (MCC) . . . 21
Operations & measurement . . . 21
Boundaries (what DO is *not*) . . . 21
8. Deliverables Summary — Windows Delivery Optimization 21

What you receive 21
Scope boundary 22
Outcome 22
Appendix A — PowerShell Scripts 23

Executive Summary — Windows Delivery Optimization (Bandwidth Savings & Peer Content Distribution)

Every managed Windows device downloads a great deal of content from Microsoft: monthly quality updates and annual feature updates, Microsoft Store apps, Microsoft 365 Apps updates, **Intune Win32 apps**, and **Autopilot / Enrollment Status Page** payloads during provisioning. Multiplied across a fleet — and especially across branch offices on modest internet circuits — that traffic saturates WAN links, slows provisioning, and drives up bandwidth cost. **Delivery Optimization (DO)** is the built-in Windows content-delivery engine that fixes this: devices share already-downloaded content with peers on the same network and pull from an on-network cache, so the same bytes cross the internet **once** instead of once per device.

This solution configures Delivery Optimization the repeatable way — through the **Intune Settings Catalog** — so a novice administrator can turn a fleet of devices that each download everything independently into a **coordinated, bandwidth-efficient** estate. It is deliberately scoped to the **content-delivery / bandwidth layer**: *how* content reaches devices efficiently. It does **not** decide *what* patches deploy or *when* (that is Windows Update for Business), how third-party apps are patched, or how Win32 apps are packaged — DO optimizes the delivery *underneath* all of those.

What you get: a click-by-click implementation that sets the correct **Download Mode**, scopes peer groups accurately across sites and subnets with **GroupIDSource**, applies sensible **bandwidth throttling** (absolute and percentage, business-hours-aware) and **cache** sizing, assigned **pilot** → **production**; a decision-level overview of **Microsoft Connected Cache (MCC)** so you know when your org needs an on-network cache node (the full MCC standup is a separate future engagement); and **monitoring** that proves the savings using the built-in `Get-DeliveryOptimizationStatus` / `Get-DeliveryOptimizationPerfSnap` cmdlets and the Intune Delivery Optimization report.

Outcome: less WAN and internet bandwidth consumed for the same updates and apps, faster branch-office provisioning, and hard numbers — peer-vs-HTTP-vs-cache bytes — that demonstrate the saving to the business. Deployable by a single administrator, validated on a pilot before you expand.

Effort model: *Assess → Configure → Pilot → Validate → Expand.*

1. Solution Overview — Windows Delivery Optimization (Bandwidth Savings & Peer Content Distribution)

What Delivery Optimization is

Delivery Optimization (DO) is the **content-delivery engine built into Windows**. When a device needs content from Microsoft, DO can source those bytes from **three** places in parallel — the Microsoft CDN / original HTTP source, an **on-network cache** (Microsoft Connected Cache), and **peer devices** that already have the content — then assembles the file from whichever sources respond fastest. The result is that a given update or app crosses the internet **once per network** rather than once per device, cutting WAN and internet bandwidth while keeping downloads fast and reliable.

This solution is the **delivery / bandwidth layer**. It does not change *what* content deploys or *when* — it makes whatever your other solutions deploy arrive **efficiently**.

Which content types Delivery Optimization can optimize

DO peer-caching and HTTP-caching apply to Microsoft-origin content, including:

- **Windows Update** — quality (monthly) updates, feature (annual) updates, and driver updates.
- **Microsoft Store** apps and app updates.
- **Microsoft 365 Apps** (Office) client updates.
- **Intune Win32 apps** (`.intunewin`) delivered by the Intune Management Extension.
- **Windows Autopilot / Enrollment Status Page (ESP)** content during provisioning.
- **Microsoft Defender** security-intelligence (definition) updates.

> *Content that is not Microsoft-origin (e.g. a third-party vendor's own download server) is not peer-cached by DO. Verify the current supported-content list against Microsoft documentation as it evolves.*

How it works — the peer-to-peer + cloud/HTTP model

1. A download begins (an update, a Store or Win32 app, an M365 Apps update, or ESP content).
2. The DO client contacts the **DO cloud service** to discover eligible **peers** (other devices with the same content on the same network / group) and any configured **cache server**.
1. DO downloads **parts** of the file **in parallel** from peers, the cache server, and the HTTP source, honoring your **bandwidth** and **cache** policy, and reassembles the complete file.
1. If peers and cache cannot supply the content, DO **falls back to the HTTP source** automatically — a download never fails just because peering is unavailable.
1. Bytes served from peers or an on-network cache **do not cross the internet**, which is the saving.

Download Mode — the primary decision (values verified against Microsoft Learn)

`DODownloadMode` dictates which sources a client may use:

Mode	Value	What it does	Typical use
HTTP Only	`0`	No peering. HTTP from the CDN/original source or a Connected Cache.	Peering not wanted; still allows cache.
LAN	`1` *(default)*	Peers on the **same local network** (same public IP; same subnet by default on Win11).	Single-site / flat LANs.

Mode	Value	What it does	Typical use
Group	2	Peers across **internal subnets** within the same **group** (AD site, domain, or custom GroupID). **Recommended** for most orgs with multiple subnets/VLANs.	Multi-subnet offices, VLANs.
Internet	3	Adds **internet** peer sources.	Roaming / distributed fleets (with caution).
Simple	99	Disables DO cloud services entirely (offline); HTTP only, no peering.	Isolated / offline networks.
Bypass	100	**Deprecated in Windows 11** — bypasses DO and uses BITS. Do **not** configure on Win11; use 0 to disable peering or 99 for offline.	Legacy only.

Group mode (2) is the recommended option for most multi-subnet organizations because it lets devices in different VLANs or remote offices peer with one another — provided the **group is scoped correctly** (below).

Scoping peer groups correctly — GroupID / GroupIDSource

In **Group** mode, which devices are allowed to peer together is set by `DOGroupIDSource` (verified values):

- 1 = **AD Site** · 2 = **Authenticated domain SID** · 3 = **DHCP Option 234** ·

4 = **DNS Suffix** · 5 = **Microsoft Entra tenant ID**.

- Or set a specific custom group with `DOGroupID` (a GUID identical on every device in the group).

Scoping matters: an Entra-tenant-ID group (5) makes *every* device in the tenant one giant peer group — appropriate only if they share networks; **DHCP Option 234** (3) or **AD Site** (1) keeps peering to devices that actually sit on the same physical network, which is usually what you want. `DORestrictPeerSelectionBy` can further limit peers to the same **subnet** (1) or use **Local Peer Discovery / DNS-SD** (2).

Bandwidth controls (verified names)

- **Percentage throttling:** `DOPercentageMaxForegroundBandwidth` / `DOPercentageMaxBackgroundBandwidth`

(default `0` = dynamic).

- **Absolute throttling (KB/s):** `DOMaxForegroundDownloadBandwidth` / `DOMaxBackgroundDownloadBandwidth`

— Microsoft recommends the **absolute** options for low-bandwidth sites.

• **Business hours:** `DOSetHoursToLimitForegroundDownloadBandwidth` / `DOSetHoursToLimitBackgroundDownloadBandwidth` limit differently during vs. outside business hours (and take precedence over the percentage settings when both are set).

• **Upload cap to internet peers:** `DOMonthlyUploadDataCap` (default **20 GB**; `0` = unlimited).

• **Minimum background QoS:** `DOMinBackgroundQoS` (default **20 MB/s**) — the floor below which DO pulls from HTTP to keep the download moving.

• **Foreground** = interactive/blocking (a user waiting on a Store app); **background** = deferrable (a monthly update). Throttle background hard, foreground gently.

Cache controls (verified names)

• `DOMaxCacheSize` — cache as a **percent** of disk (default **20%**); `DOAbsoluteMaxCacheSize` — a hard **GB** cap (overrides the percentage).

• `DOMaxCacheAge` — how long content is retained (default **259,200 s = 3 days**; `0` = unlimited).

• `DOMinFileSizeToCache` — minimum file size to peer-cache (default **50 MB**).

• `DOModifyCacheDrive` — relocate the cache (e.g. to a data drive) if the OS drive is tight.

• `DOMinRAMAllowedToPeer` (default **4 GB**) / `DOMinDiskSizeAllowedToPeer` (default **32 GB**) — hardware floors below which a device won't peer.

Microsoft Connected Cache (MCC) — decision-level only

MCC is an **on-premises or cloud cache node** that DO peers to: it downloads Microsoft content once and serves it to your devices over the LAN, so even the **first** device on a site pulls from the cache rather than the internet. DO points at it with `DOCacheHost` (static FQDN/IP list) or `DOCacheHostSource` (dynamic via **DHCP Option 235**). **When an org needs it:** large sites where the number of *distinct* files (not just peer copies) makes even one internet copy costly, or sites where peering alone leaves too much HTTP traffic. **This solution covers MCC at the decision/pointer level only** — it names the settings and tells you when to consider it. **The full MCC-node deployment (standing up and operating a cache server) is deferred to a separate future product** and is not authored here.

Monitoring & proving savings

• **On-device cmdlets:** `Get-DeliveryOptimizationStatus` (per-download source breakdown — peer vs. HTTP vs. cache bytes) and `Get-DeliveryOptimizationPerfSnap` (rolled-up performance/efficiency).

• **Intune report:** *Reports → Delivery Optimization* aggregates bandwidth-saving across the fleet.

• The saving is read as **BytesFromPeers + BytesFromCacheServer** relative to **BytesFromHttp**.

Benefits

- **Bandwidth & cost** — the same update/app crosses the internet once per network, not once per device.
- **Faster provisioning** — Autopilot/ESP and app installs pull from peers/cache instead of a congested WAN.
- **No new infrastructure required** to start — peer caching is built into Windows; MCC is optional.
- **Measurable** — peer/cache/HTTP byte counts prove the saving to the business.
- **Safe** — automatic HTTP fallback means a download never fails when peers/cache are unavailable.

Limitations (be honest with buyers)

- DO optimizes **delivery**, not **what/when**: it does not choose or schedule updates (that is WUfB), patch third-party apps, or package Win32 apps.
- Peering only helps where devices **share a network**; fully-remote single-user devices see little peer benefit.
- Peering across subnets requires **Group** mode with a correctly-scoped group; a wrong GroupIDSource either isolates devices or over-broadens the group.
- **Firewall/port** blocks (peer listening port **TCP 7680**) stop peering silently — devices fall back to HTTP.
- Full first-copy elimination on a site needs **MCC** (a separate engagement), not peering alone.
- **Bypass (100)** is deprecated on Windows 11 — don't use it.

The standard deployment model

> ***Assess the network and content sources → configure the DO Settings Catalog policy (download mode, group, bandwidth, cache) → assign to a Pilot → validate peering + savings with the cmdlets/report → expand to Production.*** *The repeatable way:* ***Assess → Configure → Pilot → Validate → Expand.***

2. Requirements Guide — Windows Delivery Optimization

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-intune-solutions.md`.

2.1 Licensing (Intune)

Description: An Intune-bearing license (Intune Plan 1, in M365 E3/E5 or Business Premium). Delivery Optimization itself is a **free, built-in Windows feature** — no separate DO license. **Why:** Intune delivers the DO Settings Catalog policy and the Delivery Optimization report. **Verify:** M365 admin center → **Billing** → **Licenses**; Intune portal loads. **Implement:** Assign M365 E3/E5 or Business Premium (Intune included).

2.2 Supported Windows edition and version

Description: Windows 10 (1809+) or Windows 11 — **Pro, Enterprise, or Education**. **Why:** DO peer caching and the newer settings (GroupIDSource, absolute bandwidth, business-hours) require a supported build; **Bypass mode is deprecated on Windows 11**. Home edition supports only consumer peering, not enterprise DO policy. **Verify:** `winver`; run `DELOPT-Preflight.ps1` (reports edition + build + the DO service state). **Implement:** Upgrade to a supported build/edition; enroll the device in Intune.

2.3 Delivery Optimization service running (DoSvc)

Description: The **Delivery Optimization** service (`DoSvc`) must be running on each device. **Why:** `DoSvc` is the client that performs peering, caching, and reporting; if it is disabled, DO falls back to plain HTTP with no savings. **Verify:** `Get-Service DoSvc`; run `DELOPT-Preflight.ps1`. **Implement:** Set `DoSvc` startup to **Automatic (Delayed Start)** and start it; do not disable it via a "debloat" script or third-party optimizer.

2.4 Peer connectivity — firewall / port 7680

Description: DO peers listen on **TCP 7680** for inbound peer connections on the local network. **Why:** If host or network firewalls block **TCP 7680**, peers can't reach each other and every device silently falls back to HTTP — no peering, no saving. (Group/LAN peering uses **7680**; the legacy Internet download mode additionally uses **UDP 3544 / Teredo** — verify against current Microsoft documentation for your scenario.) **Verify:** From two devices on the same subnet, confirm **TCP 7680** is reachable between them; run `DELOPT-Get-Status.ps1` after a download and check for non-zero **BytesFromPeers**. **Implement:** Allow inbound **TCP 7680** on the Windows Firewall Domain/Private profiles and on any network firewall between peering devices; do not block it on intra-site links.

2.5 Network access to Delivery Optimization cloud + content endpoints

Description: Outbound HTTPS to the DO cloud service and Microsoft content endpoints. **Why:** DO uses the cloud service to discover peers and cache hosts, and pulls HTTP content/fallback from Microsoft CDNs. Blocking these breaks peer discovery and downloads. **Verify:** From a device, the DO endpoints resolve and are reachable on 80/443. Representative hosts include `*.do.dsp.mp.microsoft.com`, `*.dl.delivery.mp.microsoft.com`, `*.emdl.ws.microsoft.com`, `*.prod.do.dsp.mp.microsoft.com` — **verify the current endpoint list against Microsoft documentation**, as it changes. **Implement:** Allow the Delivery Optimization and Windows Update URL/IP ranges through the proxy/firewall; where a proxy is present, ensure it does not strip the DO range/HTTP headers.

2.6 A single, non-conflicting delivery configuration

Description: DO policy is set from **one** authority — this Intune Settings Catalog policy — not a leftover GPO, a ConfigMgr client setting, or a manual registry hack. **Why:** Conflicting DO settings from multiple sources produce unpredictable download modes and undo the group scoping. **Verify:** Run `DELOPT-Get-Policy.ps1` (exports the effective Intune DO policy + assignments) and `DELOPT-Preflight.ps1` (reads the applied on-device config); reconcile against any GPO. **Implement:** Remove/disable conflicting GPO DO settings; if co-managed with ConfigMgr, decide which authority owns

client settings and align them.

2.7 Pilot and Production device groups

Description: Two Microsoft Entra security groups: `DELOPT - Devices - Pilot` and `DELOPT - Devices - Production`. **Why:** DO is rolled out pilot-first so peering and savings are validated on a small, same-network set before the fleet. **Verify:** Entra → **Groups**; both groups exist with the right members. **Implement:** Entra → **Groups** → **New group** → **Security**; add a handful of same-site devices to Pilot.

2.8 (Recommended) A Delivery Optimization measurement baseline

Description: A "before" reading of peer/HTTP/cache bytes on the pilot devices. **Why:** You can only prove savings against a baseline. **Verify:** Run `DELOPT-Get-Status.ps1` / `DELOPT-Report-Savings.ps1` before enabling Group mode. **Implement:** Capture the baseline during the Assess phase and keep it for the after-comparison.

Checklist

- Intune licensing assigned
- Windows 10 1809+/11 Pro/Ent/Edu — `DELOPT-Preflight.ps1`
- Delivery Optimization service (`DoSvc`) running
- Peer port **TCP 7680** allowed between same-network devices
- DO cloud + content endpoints reachable (verify current list)
- No conflicting GPO/ConfigMgr DO settings — `DELOPT-Get-Policy.ps1`
- `DELOPT - Devices - Pilot` and `DELOPT - Devices - Production` groups created
- Baseline peer/HTTP/cache reading captured

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — Windows Delivery Optimization via the Intune Settings Catalog

Follow these steps **in order**. All Delivery Optimization settings live in the Intune **Settings Catalog** under the category **Delivery Optimization**. You will build **one** policy, assign it to a **Pilot** group, validate peering and savings, then expand to **Production**. Commercial portal is `https://intune.microsoft.com`; US Government is `https://intune.microsoft.us` (see §3.9).

> ***Naming convention used throughout (reuse these names verbatim):*** *Policy:* `DELOPT - Policy - Pilot` → *later cloned to* `DELOPT - Policy - Production`. *Groups:* `DELOPT - Devices - Pilot`, `DELOPT - Devices - Production` *(from §2.7).*

Step 1 — Assess: capture the baseline

1. On two or three pilot devices on the **same subnet**, run `DELOPT-Preflight.ps1` (confirms edition/build, `DoSvc` running, port 7680) and `DELOPT-Get-Status.ps1` (records current peer/HTTP/cache bytes). Keep these readings — they are your "before".
1. Note the **network topology**: how many sites/subnets, whether devices share a public IP, and whether an on-network cache (MCC) already exists. This drives the **Download Mode** and **group** choice in Step 3.

Step 2 — Create the Delivery Optimization policy

1. In the Intune admin center, go to **Devices** → **Configuration** → **Create** → **New policy**.
2. **Platform: Windows 10 and later**. **Profile type: Settings catalog**. Click **Create**.
3. **Basics:** Name `DELOPT - Policy - Pilot`; Description *"Delivery Optimization — bandwidth savings & peer content distribution (pilot)."* Click **Next**.
1. On **Configuration settings**, click **+ Add settings**, and in the settings picker search **Delivery Optimization**. Select the **Delivery Optimization** category and add the settings in Steps 3–6 below.

Step 3 — Set the Download Mode and scope the peer group

1. Add **Download Mode** (`DODownloadMode`). For a multi-subnet organization set it to **Group (2)** — the recommended mode; devices peer across VLANs/subnets within the same group. (For a single flat LAN, **LAN (1)** is sufficient; to disable peering but keep cache use, **HTTP Only (0)**.)
1. Add **Select the source of Group IDs** (`DOGroupIDSource`) and choose the source that matches how your networks are actually segmented:

- **DHCP Option 234 (3)** — recommended when your DHCP scopes map to physical sites (each site's DHCP returns a distinct GUID → peer group = site). *Requires configuring DHCP Option 234 on each scope with a GUID — a network-team task; flag it if DHCP is not owned by you.*
- **AD Site (1)** — for domain-joined/hybrid devices whose AD sites already model the physical topology.
- **Microsoft Entra tenant ID (5)** — simplest, but makes the **entire tenant** one peer group; use only if that is genuinely what you want (small org, shared networks).

1. *(Optional, tighten peering)* Add **Select a method to restrict peer selection**

(`DORestrictPeerSelectionBy`) = **Subnet (1)** to keep peers within a subnet, or **Local Peer Discovery / DNS-SD (2)** on Windows 11 for local-network discovery without the cloud service.

1. *(Optional, custom group)* Instead of a source, set a fixed **Group ID** (`DOGroupID`) to a GUID that

is **identical on every device** meant to peer together (generate once with `[guid]::NewGuid()`). Do **not** set both a custom GroupID and rely on GroupIDSource for the same devices.

Step 4 — Apply bandwidth controls

Choose **absolute** limits for low-bandwidth branch sites (Microsoft's recommendation), or **percentage** for headroom-based control. Do not set both absolute and percentage for the same direction.

1. Add **Maximum background download bandwidth (in KB/s)** (`DOMaxBackgroundDownloadBandwidth`) and set

a value that protects the WAN for **background** (deferrable) content — e.g. a conservative cap on a small branch circuit. Background = monthly updates and similar.

1. Add **Maximum foreground download bandwidth (in KB/s)** (`DOMaxForegroundDownloadBandwidth`) and set

a **higher/looser** value (or leave `0` = dynamic) so **foreground** (a user waiting on a Store or Win32 app) stays responsive.

1. *(Alternative to absolute)* **Maximum background download bandwidth (percentage)**

(`DOPercentageMaxBackgroundBandwidth`) and **…foreground (percentage)** (`DOPercentageMaxForegroundBandwidth`), `0–100`.

1. *(Business-hours shaping)* Add **Set hours to limit background download bandwidth**

(`DOSetHoursToLimitBackgroundDownloadBandwidth`) to throttle harder **during** business hours and relax **outside** them (foreground equivalent: `DOSetHoursToLimitForegroundDownloadBandwidth`). When set, these take precedence over the percentage values.

1. Add **Monthly upload data cap** (`DOMonthlyUploadDataCap`) — leave the **20 GB** default for most,

or lower it where upload to internet peers must be constrained (`0` = unlimited). *(Uploads to internet peers only apply when Download Mode is **Internet (3)**.)*

1. *(Optional)* **Minimum background QoS** (`DOMinBackgroundQoS`, default **20 MB/s**) — the speed floor

below which DO supplements peers with HTTP so a background download still completes promptly.

Step 5 — Size the peer cache

1. Add **Max cache size** (`DOMaxCacheSize`) — percent of disk, default **20%** — or **Absolute max

cache size (in GBs) (`DOAbsoluteMaxCacheSize`) for a hard GB cap (the GB cap** overrides** the percentage). A larger cache means more content is available to peers.

1. Add **Max cache age** (`DOMaxCacheAge`) — seconds content is retained (default **259,200 = 3 days**;

set higher, or `0` = unlimited, so peers don't re-download recurring content).

1. *(Optional)* **Minimum peer caching content file size** (`DOMinFileSizeToCache`, default **50 MB**) —

lower it if you want smaller apps to peer-cache.

1. *(Optional)* **Modify cache drive** (`DOModifyCacheDrive`) — relocate the cache to a data drive (e.g.

`D:`) if the OS drive is tight.

1. *(Optional, hardware floors)* **Minimum RAM allowed to peer** (`DOMinRAMAllowedToPeer`, default **4

GB**) **and** Minimum disk size allowed to peer **(**`DOMinDiskSizeAllowedToPeer`**, default** 32 GB**).

Step 6 — (Decision-level) point at Microsoft Connected Cache, if you have one

Only if an MCC node already exists on your network — standing up MCC is a separate future engagement, not part of this deployment:

1. Add **Cache server hostname** (`DOCacheHost`) = the MCC node's FQDN/IP (comma-separate multiple), **or**
2. Add **Cache server hostname source** (`DOCacheHostSource`) = **DHCP Option 235 (1)**, or **Force (2)**

to override a static `DOCacheHost`. *(Requires DHCP Option 235 configured with the cache host — a network-team task.)*

1. Leave both **unset** if you have no cache node; DO peering + HTTP still work fully without MCC.

Step 7 — Finish and assign to Pilot

1. Click **Next** through **Scope tags** (add one if your org uses them).
2. On **Assignments**, under **Included groups** add `DELOPT - Devices - Pilot`. Click **Next**.
3. Review and click **Create**.
4. Trigger a sync (**Devices** → **Sync**, or on a device *Settings → Accounts → Access work or school → Info → Sync*) and allow policy to apply.

Step 8 — Validate on the Pilot (gate before expanding)

1. Run `DELOPT-Get-Policy.ps1` to confirm the policy and its **Pilot** assignment exist in Intune.
2. On two pilot devices on the **same subnet**, run `DELOPT-Preflight.ps1` to confirm the applied Download Mode/Group.
1. Trigger real content on **device A** (e.g. a Store app update or a Win32 app install), let it complete, then start the **same** content on **device B**. Run `DELOPT-Get-Status.ps1` on device B — **BytesFromPeers** should be **non-zero** (B pulled from A).

1. Run `DELOPT-Report-Savings.ps1` across the pilot set and compare peer/cache vs. HTTP bytes to

your Step 1 baseline. **Do not expand until peering is proven** (see the pass/fail gates in `05-testing.md`).

Step 9 — Expand to Production

1. In Intune, open `DELOPT - Policy - Pilot`, use **Duplicate**, and name the copy

`DELOPT - Policy - Production` (or simply add the Production group to the existing policy).

1. Reconcile any values you adjusted during the pilot (bandwidth caps, cache size, group source).

2. On **Assignments**, add `DELOPT - Devices - Production` (and, when confident, remove the Pilot

from the pilot-only policy to avoid double assignment).

1. Roll out in waves; after each wave re-run `DELOPT-Report-Savings.ps1` to confirm savings scale.

3.9 US Government (GCC / GCC High / DoD) notes

• Use the **US Gov portals**: Intune `https://intune.microsoft.us`, Entra `https://portal.azure.us`,

and Microsoft Graph `https://graph.microsoft.us` (scripts: `Connect-MgGraph -Environment USGov`, or `USGovDoD` for DoD).

• The **Settings Catalog Delivery Optimization settings are the same**; the setting names/values in

Steps 3–6 are identical. The difference is the **admin endpoint** you configure them from.

• **DO/MCC cloud-endpoint reachability may differ** in sovereign clouds — the DO cloud service and

content CDNs use US-government service endpoints; confirm the current endpoint list and that your proxy/firewall allows them. **Verify for GCC High** before assuming parity of any specific endpoint.

5. Testing & Validation — Delivery Optimization

Validate on the **Pilot** before expanding. Each test has an explicit **PASS/FAIL gate**. Do not move to Production until every gate passes on same-network pilot devices.

Test 1 — Policy applied

Procedure: Run `DELOPT-Get-Policy.ps1` (confirms the Intune policy + Pilot assignment) and, on a pilot device, `DELOPT-Preflight.ps1` (reads the applied on-device DO config). **PASS:** `DELOPT - Policy - Pilot` exists, is assigned to `DELOPT - Devices - Pilot`, and the device reports the expected **Download Mode** (e.g. Group/2) and Group source. **FAIL:** Policy missing/unassigned, or the device still shows the default LAN(1)/unconfigured mode → re-check assignment and sync; see `06-troubleshooting.md`.

Test 2 — Delivery Optimization service healthy

Procedure: On a pilot device, `Get-Service DoSvc`; run `DELOPT-Preflight.ps1`. **PASS:** `DoSvc` is **Running** and startup is Automatic (Delayed Start). **FAIL:** `DoSvc` stopped/disabled → DO can't peer; re-enable and start it.

Test 3 — Peer connectivity (port 7680)

Procedure: From device B, test **TCP 7680** to device A on the same subnet (`Test-NetConnection <A> -Port 7680`). **PASS:** Port 7680 is open between the two devices. **FAIL:** Port filtered/closed → peering will not occur; open **TCP 7680** on host/network firewalls.

Test 4 — Peering actually happens (the core proof)

Procedure: Fully download a piece of Microsoft content on **device A** (a Store app update, a Win32 app, or a monthly update). Then initiate the **same** content on **device B** on the same subnet. Run `DELOPT-Get-Status.ps1` on device B. **PASS:** Device B shows **non-zero BytesFromPeers** (and/or BytesFromCacheServer if MCC is present) for that download — content came from A, not the internet. **FAIL:** BytesFromPeers = 0 and all bytes are BytesFromHttp → check Download Mode/group scoping, port 7680, and that both devices are in the same group; see `06-troubleshooting.md`.

Test 5 — Group scoping is correct

Procedure: On device B, run `DELOPT-Get-Status.ps1`; confirm peers are only devices that *should* be in the group (same site/subnet, per your GroupIDSource). **PASS:** Peers are same-network devices only; devices in unrelated sites are **not** peering. **FAIL:** Devices across the whole tenant are peering (over-broad, e.g. GroupIDSource=Entra tenant when you wanted per-site), or same-site devices are **not** peering (over-narrow) → adjust `DOGroupIDSource`.

Test 6 — Bandwidth caps respected

Procedure: During a large background download, observe the DO download rate (Task Manager / Resource Monitor, or `Get-DeliveryOptimizationStatus` download rate) against your configured `DOMaxBackgroundDownloadBandwidth`. **PASS:** Background download rate stays at/under the configured cap; foreground content remains responsive. **FAIL:** Rate exceeds the cap or foreground is starved → re-check that only one of absolute/percentage is set per direction, and business-hours precedence.

Test 7 — Savings measured vs. baseline

Procedure: Run `DELOPT-Report-Savings.ps1` across the pilot set and compare peer + cache bytes vs. HTTP bytes against the Step-1 baseline. **PASS:** A meaningful share of bytes now comes from **peers/cache** (a positive saving vs. baseline). **FAIL:** No measurable shift from HTTP to peers/cache → peering isn't working (revisit Tests 3–5) or there isn't enough shared content yet (repeat after more devices have common content).

Test 8 — HTTP fallback safety

Procedure: On an **isolated** pilot device (no peers available), start a Microsoft download. **PASS:** The download completes over HTTP — DO never blocks a download when peers/cache are unavailable. **FAIL:** Download stalls → this is not expected for modes 0/1/2/3; check for Bypass(100) misconfig or a broken proxy path.

Exit criteria (all must be true before Production)

- Policy applied with the intended Download Mode + group (Test 1)
- `DoSvc` running (Test 2) and port 7680 open between peers (Test 3)
- **BytesFromPeers non-zero** on a second device for shared content (Test 4)
- Peer group scoped correctly — right devices only (Test 5)
- Bandwidth caps honored; foreground responsive (Test 6)
- Measured peer/cache saving vs. baseline (Test 7)
- HTTP fallback proven safe (Test 8)

6. Troubleshooting — Delivery Optimization

Symptom → likely cause → fix. Run `DELOPT-Preflight.ps1` and `DELOPT-Get-Status.ps1` first to gather the on-device facts.

No peering — every device downloads over HTTP (BytesFromPeers = 0)

Likely causes & fixes:

- **Port 7680 blocked.** Host or network firewall blocks the peer listening port. Test `Test-NetConnection <peer> -Port 7680`; allow **inbound TCP 7680** on the Domain/Private profile and on intra-site network firewalls.
- **Wrong Download Mode.** Devices are on **HTTP Only (0)** or **Simple (99)**, or on **LAN (1)** across different subnets. Set **Group (2)** (with a group source) to peer across subnets; confirm with `DELOPT-Preflight.ps1`.
- `DoSvc` **not running.** The Delivery Optimization service is stopped/disabled (often by a "debloat" script). Set it to **Automatic (Delayed Start)** and start it.
- **No shared content yet.** Peers can only share content they already have. Ensure device A fully downloaded the item before device B requests the **same** item; retry Test 4.
- **DO cloud service unreachable.** Peer discovery needs the DO cloud endpoints on 80/443. Verify the endpoint list (it changes) and that the proxy isn't stripping them.

Devices peer with the wrong set (group scoped incorrectly)

- **Over-broad (whole tenant peering):** `DOGroupIDSource` = **Entra tenant ID (5)** makes every device

one group. Switch to **DHCP Option 234 (3)** or **AD Site (1)** for per-site peering.

- **Over-narrow (same-site devices don't peer):** `DORestrictPeerSelectionBy` = **Subnet (1)** on a site

with several subnets, or a mismatched custom `DOGroupID`. Loosen to the site level or fix the GroupID so it is identical on all intended peers.

- **DHCP Option 234 not set / wrong:** Group source is DHCP but the scope has no (or a malformed) Option

234 GUID → DO falls back to default grouping. Have the network team set Option 234 correctly per scope.

Savings don't appear in reports

- **Baseline missing or too early.** Compare against the Step-1 baseline; savings only show once devices

share content. Re-run `DELOPT-Report-Savings.ps1` after more devices have common items.

- **Intune Delivery Optimization report lag/data.** The report needs devices reporting in; allow time

and confirm devices are checking in. Cross-check with on-device `Get-DeliveryOptimizationStatus`.

- **Peering actually failing.** If on-device BytesFromPeers is 0, the "no peering" section above is the

real problem — reports can't show a saving that isn't happening.

Conflicting configuration / policy not applying

- **Two authorities.** A leftover **GPO** or **ConfigMgr** client setting is fighting the Intune policy.

Run `DELOPT-Get-Policy.ps1` (Intune side) and compare to on-device config; remove conflicting GPO DO settings or align co-management.

- **Assignment/sync.** Policy exists but the device shows defaults → confirm the device is in

`DELOPT - Devices - Pilot`, force a sync, and re-check. Settings Catalog conflicts (same setting set by two profiles) show as an error in the device's policy status — resolve the duplicate.

Bandwidth caps not respected / foreground starved

- **Both absolute and percentage set** for the same direction, or **business-hours** settings overriding

as designed. Set only one of absolute/percentage per direction; remember business-hours settings take precedence over the percentage values.

- **Foreground throttled too hard.** Raise `DOMaxForegroundDownloadBandwidth` (or leave `0` = dynamic) so

interactive content stays responsive; keep the hard cap on **background** only.

Bypass (100) side-effects on Windows 11

- **Content fails to download / DO bypassed.** `DODownloadMode` = **100 (Bypass)** is **deprecated on

Windows 11 **and routes around DO via BITS. Change to** 0 **(disable peering) or** 99** (offline); do not use 100 on Win11.

US Government specifics

- **Endpoints unreachable in GCC High / DoD.** Sovereign clouds use US-government DO/content endpoints.

Verify the current endpoint list and allow the correct US-gov ranges; configure policy from `https://intune.microsoft.us` and Graph via `Connect-MgGraph -Environment USGov` (or `USGovDoD`). **Verify for GCC High** where a specific endpoint's parity is uncertain.

7. Best Practices — Delivery Optimization

Download mode & grouping

- **Prefer Group (2)** for any organization with more than one subnet/VLAN — it is the mode that

produces the biggest bandwidth saving by letting devices peer across internal networks.

- **Scope the group to the physical network, not the whole tenant.** Match `DOGroupIDSource` to how

your networks actually segment: **DHCP Option 234 (3)** or **AD Site (1)** for per-site peering; reserve **Entra tenant ID (5)** for small orgs where all devices genuinely share networks.

- **Don't over-narrow either** — restricting to subnet when a site has several small subnets can strand

devices with no peers. Validate real peering (Test 4) before committing.

- **Never use Bypass (100) on Windows 11** — it is deprecated. Use **HTTP Only (0)** to disable peering

or **Simple (99)** for offline networks.

Bandwidth

- **Use absolute (KB/s) caps on low-bandwidth branch circuits** — Microsoft's own recommendation — and

reserve percentage caps for well-provisioned sites.

- **Throttle background hard, foreground gently.** Users wait on foreground (interactive) content;

monthly updates are background and can be squeezed.

- **Use business-hours limits** to protect the WAN during the workday and open the taps after hours.

- **Leave the 20 GB monthly upload cap** unless you have a specific reason to lower it; uploads to

internet peers only apply in Internet (3) mode.

Cache

• **Give the cache room.** A bigger, longer-lived cache (`DOMaxCacheSize`/`DOAbsoluteMaxCacheSize`, `DOMaxCacheAge`) keeps more content available for peers, which raises the peer hit-rate and the saving.

• **Relocate the cache** (`DOModifyCacheDrive`) on devices with small OS drives so DO isn't starved.

Microsoft Connected Cache (MCC)

• **Consider MCC when peering alone leaves too much HTTP traffic** — large sites, or sites where the *first* copy of every distinct file still crosses the internet. MCC eliminates that first copy.

• **Keep MCC as a separate project.** Standing up and operating a cache node is its own engagement; this solution only points DO at an existing node (`DOCacheHost` / `DOCacheHostSource`).

Operations & measurement

• **Baseline before, measure after.** Always capture peer/HTTP/cache bytes before enabling Group mode and after, so the saving is a number, not a claim (`DELOPT-Report-Savings.ps1`).

• **Watch the Intune Delivery Optimization report** for fleet-wide trends after each rollout wave.

• **Roll out in waves** and re-measure — savings compound as more devices share content.

• **Keep one authority.** Set DO only from this Intune policy; remove conflicting GPO/ConfigMgr DO settings so group scoping stays intact.

Boundaries (what DO is *not*)

• DO does **not** choose or schedule updates — that is **Windows Update for Business** (update rings / what & when).

• DO does **not** patch third-party apps (**Third-Party App Patching**) or package Win32 apps (**Win32 App Packaging & Deployment**).

• DO optimizes the **delivery** of all of the above. Deploy those solutions for the *what/when*; deploy this one so it all arrives **efficiently**.

8. Deliverables Summary — Windows Delivery Optimization

What you receive

• **Implementation Guide (PDF + Word)** — the complete, click-by-click solution:

• Solution Overview — the DO peer + cloud/HTTP model, which content types it optimizes, Download Mode values, group scoping, bandwidth and cache controls, MCC at decision level.

• Requirements Guide — edition/version, `DoSvc`, port 7680, endpoints, single-authority, groups —

each with Verify / Implement-if-missing.

• Implementation Guide — build the DO Settings Catalog policy step by step (download mode, group, bandwidth, cache), assign pilot → production, including US Government portal notes.

• Testing & Validation — 8 tests with PASS/FAIL gates proving peering and measured savings.

• Troubleshooting — no peering, wrong group, firewall/port blocks, savings not appearing.

• Best Practices — mode/group, bandwidth, cache, MCC, and measurement discipline.

• **5 PowerShell scripts (ZIP)** — read-only by default; remediation is guarded with `-WhatIf`:

• `DELOPT-Preflight.ps1` — verify Windows edition/version, `DoSvc`, and peer port 7680.

• `DELOPT-Get-Status.ps1` — per-download source breakdown (peer / HTTP / cache bytes) via `Get-DeliveryOptimizationStatus` + `Get-DeliveryOptimizationPerfSnap`.

• `DELOPT-Report-Savings.ps1` — aggregate savings across a set of devices.

• `DELOPT-Get-Policy.ps1` — export the Intune DO config policy + assignments (Microsoft Graph, cloud-aware: Global / USGov / USGovDoD).

• `DELOPT-Remediation.ps1` — guarded (`-WhatIf` default) fix for a missing/incorrect DO setting + service/port checks.

Scope boundary

This solution is the **content-delivery / bandwidth** layer only. It does not choose or schedule updates (Windows Update for Business), patch third-party apps (Third-Party App Patching), or package Win32 apps (Win32 App Packaging & Deployment). The **full Microsoft Connected Cache node deployment** is deferred to a separate future product; here MCC is covered at decision/pointer level only.

Outcome

The same updates and apps cross the internet **once per network** instead of once per device — lower WAN/internet bandwidth, faster provisioning, and measured peer/cache-vs-HTTP savings you can show the business. Deployable by a single administrator; validated on a pilot before you expand.

repeatable model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

DELOPT-Get-Policy.ps1

Purpose: Finds Settings Catalog configuration policies that contain Delivery Optimization settings in Intune and lists their assigned groups, so you can confirm the DELOPT policy exists and is assigned to the intended Pilot/Production groups (and spot conflicting duplicates). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\DELOPT-Get-Policy.ps1 Sovereign cloud: .\DELOPT-Get-Policy.ps1 -Environment USGov (or USGovDoD) Requires the Microsoft.Graph SDK and delegated DeviceManagementConfiguration.Read.All. EXPECTED RESULTS A list of Delivery Optimization configuration policies with their assigned group IDs. Read-only against Graph (no changes made). .NOTES White-label — Intune Solutions. Read-only against Microsoft Graph. > [CmdletBinding()] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global')

How to run: powershell -ExecutionPolicy Bypass -File .\DELOPT-Get-Policy.ps1 Sovereign cloud: .\DELOPT-Get-Policy.ps1 -Environment USGov (or USGovDoD) Requires the Microsoft.Graph SDK and delegated DeviceManagementConfiguration.Read.All. EXPECTED RESULTS A list of Delivery Optimization configuration policies with their assigned group IDs. Read-only against Graph (no changes made). .NOTES White-label — Intune Solutions. Read-only against Microsoft Graph. > [CmdletBinding()] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global')

DELOPT-Get-Status.ps1

Purpose: Reads the local Delivery Optimization activity with Get-DeliveryOptimizationStatus and the rolled-up efficiency from Get-DeliveryOptimizationPerfSnap, and summarizes how many bytes came from peers vs. an HTTP source vs. a cache server (MCC) — the direct evidence that peering works. HOW TO RUN Elevated: powershell -ExecutionPolicy Bypass -File .\DELOPT-Get-Status.ps1 EXPECTED RESULTS Per-download rows (file, status, BytesFromPeers / BytesFromHttp / BytesFromCacheServer) and a total peer-vs-HTTP-vs-cache summary. Non-zero BytesFromPeers proves peering. Read-only. .NOTES White-label — Intune Solutions. Read-only (local DO cmdlets, no changes made). > [CmdletBinding()] param()

How to run: Elevated: powershell -ExecutionPolicy Bypass -File .\DELOPT-Get-Status.ps1 EXPECTED RESULTS Per-download rows (file, status, BytesFromPeers / BytesFromHttp / BytesFromCacheServer) and a total peer-vs-HTTP-vs-cache summary. Non-zero BytesFromPeers proves peering. Read-only. .NOTES White-label — Intune Solutions. Read-only (local DO cmdlets, no changes made). > [CmdletBinding()] param()

DELOPT-Preflight.ps1

Purpose: Confirms the device can participate in Delivery Optimization: supported Windows edition/version, the Delivery Optimization service (DoSvc) running, the peer listening port (TCP 7680) open locally, and the effective on-device DO Download Mode / group config. HOW TO RUN Elevated: powershell -ExecutionPolicy Bypass -File .\DELOPT-Preflight.ps1 EXPECTED RESULTS Edition/build, DoSvc state, port 7680 listening status, and the applied Download Mode / GroupID source — with a PASS/REVIEW verdict per check. Read-only. .NOTES White-label — Intune Solutions. Read-only (no changes made). > [CmdletBinding()] param()

How to run: Elevated: powershell -ExecutionPolicy Bypass -File .\DELOPT-Preflight.ps1 EXPECTED RESULTS Edition/build, DoSvc state, port 7680 listening status, and the applied Download Mode / GroupID source — with a PASS/REVIEW verdict per check. Read-only. .NOTES White-label — Intune Solutions. Read-only (no changes made). > [CmdletBinding()] param()

DELOPT-Remediation.ps1

Purpose: Safely corrects the most common local blockers that stop peering: the Delivery Optimization service (DoSvc) not running, and the peer listening port (TCP 7680) not allowed on the local Windows Firewall. It can also stamp a Download Mode into the DO policy registry as a stop-gap where an Intune policy is not yet applied. Defaults to -WhatIf (shows actions, changes nothing). HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\DELOPT-Remediation.ps1 Apply: .\DELOPT-Remediation.ps1 -Apply Optional: -SetDownloadMode 2 (only as a local stop-gap; the Intune policy is authoritative) EXPECTED RESULTS A list of intended actions in preview; with -Apply, DoSvc set Automatic + started, an inbound allow rule for TCP 7680, and (if requested) the DODownloadMode value written. .NOTES White-label — Intune Solutions. Guarded: no changes unless -Apply is passed. Run elevated to change service/firewall/registry. The Intune Settings Catalog policy is the authoritative configuration. > [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='High')] param([switch] $Apply, [ValidateSet(0,1,2,3,99)] [int] $SetDownloadMode = -1)

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\DELOPT-Remediation.ps1 Apply: .\DELOPT-Remediation.ps1 -Apply Optional: -SetDownloadMode 2 (only as a local stop-gap; the Intune policy is authoritative) EXPECTED RESULTS A list of intended actions in preview; with -Apply, DoSvc set Automatic + started, an inbound allow rule for TCP 7680, and (if requested) the DODownloadMode value written. .NOTES White-label — Intune Solutions. Guarded: no changes unless -Apply is passed. Run elevated to change service/firewall/registry. The Intune Settings Catalog policy is the authoritative configuration. > [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='High')] param([switch] $Apply, [ValidateSet(0,1,2,3,99)] [int] $SetDownloadMode = -1)

DELOPT-Report-Savings.ps1

Purpose: Collects the Delivery Optimization source breakdown (peer / HTTP / cache bytes) from one or more devices and produces a combined savings report — the bandwidth NOT pulled from the internet because it came from peers or an on-network cache. Use it to prove savings vs. the pre-deployment baseline. HOW TO RUN Local only: powershell -ExecutionPolicy Bypass -File .\DELOPT-Report-Savings.ps1 Multi-device: .\DELOPT-Report-Savings.ps1 -ComputerName PC1,PC2,PC3 (uses WinRM/PS remoting) Optional: -CsvPath .\delopt-savings.csv to also write a CSV. EXPECTED RESULTS A per-device table (peer/cache/HTTP MB) plus a fleet total and the percentage of downloaded bytes saved from the internet. Read-only. .NOTES White-label — Intune Solutions. Read-only. Remoting requires WinRM enabled + rights on targets. > [CmdletBinding()] param([string[]] $ComputerName = @($env:COMPUTERNAME), [string] $CsvPath)

How to run: Local only: powershell -ExecutionPolicy Bypass -File .\DELOPT-Report-Savings.ps1 Multi-device: .\DELOPT-Report-Savings.ps1 -ComputerName PC1,PC2,PC3 (uses WinRM/PS remoting) Optional: -CsvPath .\delopt-savings.csv to also write a CSV. EXPECTED RESULTS A per-device table (peer/cache/HTTP MB) plus a fleet total and the percentage of downloaded bytes saved from the internet. Read-only. .NOTES White-label — Intune Solutions. Read-only. Remoting requires WinRM enabled + rights on targets. > [CmdletBinding()] param([string[]] $ComputerName = @($env:COMPUTERNAME), [string] $CsvPath)